CHAPTER 6

Equated Monthly Installment (EMI) & Amortization

Annuity-Immediate Valuation, Finite Geometric Series Proofs, Balance Recurrence Relations, and Prepayment Mechanics

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6.1 Definition & Axioms of Debt Amortization

An Equated Monthly Installment (EMI) is a fixed periodic payment made by a borrower to a lender to fully amortize an outstanding debt over a specified term. Each installment is partitioned into two distinct components: an **interest payment** covering the cost of unpaid capital, and a **principal repayment** that reduces the balance.

Definition 6.1 (The Closed-Form EMI Equation)

Let loan principal be $P > 0$, nominal annual interest rate be $r \geq 0$, and total tenure be t years. The periodic monthly interest rate is $r_m = r / 12$, and total payment count is $N = 12 \cdot t$.
The constant monthly installment E satisfying full amortization is:

$$E = P \cdot [\, r_m (1 + r_m)^N \,] / [\, (1 + r_m)^N - 1 \,]$$

When $r_m = 0$, the equation reduces by L'Hôpital's Rule to $E = P / N$.

6.2 First-Principles Derivation via Geometric Series

No-Arbitrage Equilibrium Proof:
By the Principle of Temporal Financial Equivalence, the borrowed principal P must equal the sum of the present values of all future N monthly installments discounted at periodic rate r_m:

- $P = \sum_{k=1}^{N} [\, E / (1 + r_m)^k \,] = E \sum_{k=1}^{N} v^k$, where $v = (1 + r_m)^{-1}$.
- The summation is a finite geometric series with initial term $a_1 = v$, ratio $q = v$, and N terms:
  $\sum_{k=1}^{N} v^k = v [\, (1 - v^N) / (1 - v) \,]$.
- Substituting $v = 1 / (1 + r_m)$:
  $1 - v = 1 - 1/(1 + r_m) = r_m / (1 + r_m) = r_m \cdot v$.
- Therefore: $\sum_{k=1}^{N} v^k = v [\, (1 - (1+r_m)^{-N}) / (r_m \cdot v) \,] = [\, 1 - (1 + r_m)^{-N} \,] / r_m$.
- Equating $P = E \cdot [\, (1 - (1+r_m)^{-N}) / r_m \,]$ and multiplying numerator and denominator by $(1+r_m)^N$ yields Definition 6.1.

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Theorem 6.1 (Amortization Balance Recurrence & Principal Growth)

For payment period $k \in \{1, 2, ..., N\}$ with opening balance B_{k-1} (where $B_0 = P$):

- **Interest Component:** $I_k = B_{k-1} \cdot r_m$
- **Principal Component:** $P_k = E - I_k$
- **Closing Balance:** $B_k = B_{k-1} - P_k = B_{k-1}(1 + r_m) - E$.

Principal Progression Property: $P_k = P_1 \cdot (1 + r_m)^{k-1}$

Analytical Consequence: Principal repayment grows exponentially over time at rate r_m, meaning early installments consist overwhelmingly of interest, while terminal installments consist almost entirely of principal.

6.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: full amortization breakdown of a 20-year mortgage, and prepayment acceleration savings analysis.

Example 6.1: 20-Year Home Loan with 3-Month Amortization Trace

Problem Statement: A borrower secures a residential mortgage of ***P* = INR 5,000,000** at an annual nominal rate ***R* = 8.50%** for tenure ***t* = 20 years** (N = 240 months). Calculate the exact monthly EMI, lifetime total interest liability, and the first 3 monthly amortization splits.

Step-by-Step Analytical Solution:

Step 1 (Parameter Normalization & Monthly Rate):

- Monthly rate: $r_m = 0.085 / 12 = 0.007083333...$
- Total payment count: $N = 20 \times 12 = 240$ months.
- Growth factor: $(1 + r_m)^{240} = (1.007083333)^{240} \approx 5.4344755$.

Step 2 (Closed-Form EMI Calculation):

- Numerator: $5{,}000{,}000 \times 0.007083333 \times 5.4344755 = 192{,}471.01$
- Denominator: $5.4344755 - 1 = 4.4344755$
- **EMI:** $E = 192{,}471.01 / 4.4344755 =$ **INR 43,407.84** per month.

Step 3 (Lifetime Interest Obligation):

- Total Outflow: $43{,}407.84 \times 240 =$ **INR 10,417,881.60**.
- Total Interest: $10{,}417{,}881.60 - 5{,}000{,}000 =$ **INR 5,417,881.60** (exceeds borrowed capital!).

Step 4 (First 3-Month Amortization Schedule Trace):

- **Month 1:** Interest $I_1 = 5{,}000{,}000 \times 0.00708333 =$ INR 35,416.67.
  Principal $P_1 = 43{,}407.84 - 35{,}416.67 =$ **INR 7,991.17**. Balance $B_1 =$ **INR 4,992,008.83**.
- **Month 2:** Interest $I_2 = 4{,}992{,}008.83 \times 0.00708333 =$ INR 35,360.06.
  Principal $P_2 = 43{,}407.84 - 35{,}360.06 =$ **INR 8,047.78**. Balance $B_2 =$ **INR 4,983,961.05**.
- **Month 3:** Interest $I_3 = 4{,}983{,}961.05 \times 0.00708333 =$ INR 35,303.06.
  Principal $P_3 = 43{,}407.84 - 35{,}303.06 =$ **INR 8,104.78**. Balance $B_3 =$ **INR 4,975,856.27**.

Example 6.2: Prepayment Acceleration & Lifetime Interest Savings

Problem Statement: On the mortgage in Example 6.1, the borrower executes a lump-sum principal prepayment of **INR 500,000 at the end of Year 1 (Month 12)**. If the EMI remains unchanged at **INR 43,407.84**, calculate the revised remaining tenure and total interest saved.

Step-by-Step Analytical Solution:

Step 1 (Balance at Month 12 Before Prepayment):

Balance after 12 payments: $B_{12} = P(1+r_m)^{12} - E\,[\,((1+r_m)^{12} - 1) / r_m\,]$.

- Compound principal: $5{,}000{,}000 \times (1.00708333)^{12} = 5{,}000{,}000 \times 1.0883909 = \text{INR } 5{,}441{,}954.50$.
- Future value of 12 EMIs: $43{,}407.84 \times (0.0883909 / 0.00708333) = 43{,}407.84 \times 12.4787 = \text{INR } 541{,}673.50$.
- Balance $B_{12} = 5{,}441{,}954.50 - 541{,}673.50 =$ **INR 4,900,281.00**.

Step 2 (Adjusting Balance Post-Prepayment):

- Revised Balance: $B'_{12} = 4{,}900{,}281.00 - 500{,}000 =$ **INR 4,400,281.00**.

Step 3 (Solving for Revised Remaining Tenure N'):

From Definition 6.1: $1 - (1 + r_m)^{-N'} = (B'_{12} \cdot r_m) / E$.

- $(4{,}400{,}281.00 \times 0.007083333) / 43{,}407.84 = 31{,}168.66 / 43{,}407.84 = 0.718042$.
- $(1.007083333)^{-N'} = 1 - 0.718042 = 0.281958$.
- $-N' \cdot \ln(1.007083333) = \ln(0.281958)$ &implies; $-N' \times 0.0070584 = -1.265997$.
- **Revised Remaining Months:** $N' = 1.265997 / 0.0070584 \approx$ **179.36 months** (≈ 180 months / 15 years).

Step 4 (Total Interest Savings Analysis):

- Original remaining payments: $240 - 12 = 228$ months.
- Months saved: $228 - 180 =$ **48 months (4 full years eliminated!)**.
- Total Outflow Saved: $48 \times 43{,}407.84 =$ **INR 2,083,576.32**.
- Net Interest Saved: 2,083,576.32 – 500,000 prepayment = **INR 1,583,576.32 pure savings**.

6.4 Algorithmic Implementation in Python

Below is the complete implementation from models/loans_and_credit.py, featuring closed-form EMI calculation and full amortization schedule generation:

Listing: models/loans_and_credit.py — Loan Amortization Engine

```
from typing import List, Dict

def calculate_emi(principal: float, annual_rate: float, tenure_years: float) -> float:
    """Computes closed-form EMI: E = P * [r_m * (1+r_m)^N] / [(1+r_m)^N - 1]."""
    if principal <= 0.0 or annual_rate < 0.0 or tenure_years <= 0.0:
        raise ValueError("Principal and tenure must be strictly positive.")
    if annual_rate == 0.0:
        return float(principal / (tenure_years * 12.0))
    monthly_rate = annual_rate / 12.0
    total_months = int(round(tenure_years * 12.0))
    growth_factor = (1.0 + monthly_rate) ** total_months
    return float(principal * (monthly_rate * growth_factor) / (growth_factor - 1.0))


def generate_amortization_schedule(principal: float, annual_rate: float, tenure_years: float) -> List[Dict]:
    """Generates detailed monthly amortization schedule with balance recurrence."""
    emi = calculate_emi(principal, annual_rate, tenure_years)
    monthly_rate = annual_rate / 12.0
    balance = principal
    schedule = []
    total_months = int(round(tenure_years * 12.0))
    for month in range(1, total_months + 1):
        interest_paid = balance * monthly_rate
        principal_paid = emi - interest_paid
        balance -= principal_paid
        schedule.append({'month': month, 'interest': interest_paid, 'principal': principal_paid, 'balance': max(0.
    return schedule
```

6.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
Annuity Class	Annuity-Immediate	Cash flows transacted at the end of each payment period
Principal Component	$P_k = P_1(1 + r_m)^{k-1}$	Principal repayment accelerates exponentially over tenure
Interest Component	$I_k = E - P_k$	Interest burden declines monotonically to zero at terminal month
Zero-Rate Limit	$\lim_{r \to 0} E = P / N$	Smooth algebraic recovery of interest-free straight amortization
Prepayment Elasticity	$\partial N / \partial B > 0$	Early balance reduction yields superlinear tenure compression